The Grand Challenge of PQC Migration: Analysis of Modern Blockchain and Intertwined Human Egoisms

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Abstract. Since the invention of Bitcoin in 2008, blockchain technology has evolved from a simple decentralized ledger into a complex, multi-functional financial system spanning multiple layers. Despite this evolution, the cryptographic primitives foundational to this trustless system are now threatened by the emergence of quantum computers. While the necessity of Post-Quantum Cryptography (PQC) migration is widely acknowledged, blockchain remains an extremely challenging domain for cryptographic migration due to its decentralized architecture and long-term security requirements. Today, the evolving complexity has rendered simple cryptographic migration models insufficient at system scale. Furthermore, human factors rooted in economic rationality and intertwined incentives, such as investors' pursuit of short-term profits and miners' mining profitability, have emerged as critical obstacles to migration. This paper provides a comprehensive analysis of the challenges and current status of blockchain PQC migration as of December 2025. Specifically, we analyze the components requiring migration and their technical challenges, while systematically organizing operational barriers, including trade-offs found within migration strategies and stakeholder interests. Furthermore, we propose guidelines for future research initiatives that academia and the blockchain community should collaboratively address. By synthesizing ongoing community discussions and existing literature, this work aims to establish a foundation for navigating the path toward achieving quantum-resistant blockchains.

Keywords: Blockchain · Post-Quantum Cryptography · PQC Migration.

1 Introduction

1.1 Motivation

Currently, any protocols relying on the security of public key cryptography are threatened to be compromised by Cryptographically Relevant Quantum Com-

puters (CRQCs). However, for many entities, it remains a reactive measure compelled by necessity rather than a proactive initiative, because the transition to Post-Quantum Cryptography (PQC) does not inherently yield direct economic benefits.

Against this backdrop, PQC migration for blockchain presents an extremely formidable challenge. The challenges stem not only from the technical difficulties associated with updating decentralized networks and the requirements of long-term security assurance, but also from the necessity of addressing the misaligned incentives and conflicting speculation among various stakeholders.

1.2 The Threat of Quantum Computers to Blockchain

Major blockchains, such as Bitcoin [29] and Ethereum [8], employ ECDSA for transaction signing. The security of this digital signature algorithm relies on the computational difficulties of solving the discrete logarithm problem. However, Shor's algorithm [37] theoretically demonstrates that the discrete logarithm problem can be solved in polynomial time, rendering ECDSA vulnerable to CRQCs. Specifically, if a sufficiently powerful quantum computer capable of executing Shor's algorithm emerges, it becomes possible to derive a user's signing key from the verification key exposed on-chain. This would allow an attacker to forge valid signatures and execute unauthorized asset transfers, collapsing the core principle of blockchain technology; "Not your keys, not your coins."

In contrast, the threat to symmetric key cryptography and hash functions is considered less severe. While Grover's algorithm [22] provides a speedup for unstructured search problems effectively halving the security level[6], security can be maintained by doubling the key or hash length. Consequently, cryptographic hash functions used in hash tree are considered to be quantum-resistant. Therefore, the primary focus of PQC research and migration efforts is directed toward the replacement of vulnerable public-key cryptography.

One example of concrete threat analysis is the study targeting Bitcoin by Giechaskiel et al. [21]. As of 2016, this study categorized attack models assuming compromised signature algorithms and hash functions and their combinations, pointing out the critical risks such as double-spending attack. However, the scope of this analysis was limited to the Bitcoin specifications of that time. Applying these findings to the currently complex blockchain environment requires careful consideration, and a comprehensive re-evaluation from a contemporary perspective. This involves taking the enhanced functionalities of smart contract platforms such as Ethereum into account and analyzing the threats to the integrity of historical data on such platforms.

1.3 Evoloving Blockchain Ecosystem and Lack of Framework

The scope of systems threatened by CRQCs has expanded dramatically compared to the nascent stages of blockchain technology. Early blockchains, as exemplified by Bitcoin, were designed primarily as peer-to-peer electronic cash

[6] For example, the preimage resistance of SHA-256 gets equivalent to 128 bits.

systems. Today, however, diverse blockchain configurations have transformed into multi-functional financial infrastructures supporting Decentralized Finance (DeFi), accompanied by the maturation of off-chain protocols and scaling solutions. Furthermore, diverse stakeholders not envisioned at first such as centralized exchanges and custodians have become indispensable to the ecosystem, thereby significantly increasing the complexity of the entire system.

Nevertheless, discussions regarding PQC migration have failed to keep pace with this rapid evolution, resulting in a fragmented discourse. Much of the existing proposals and debates focus on technical aspects of replacing signature algorithms used for transaction verification. However, this narrow view fails to capture the full extent of the modern blockchain protocol stack and much more complex ecosystem. To date, no comprehensive framework has been proposed that systematically catalogs the components requiring transition and the governance challenges among stakeholders. Moreover, even the core tenets of a migration strategy, such as the definition of the migration scope and the criteria for migration completion, remain ambiguous.

1.4 Our Contribution

To bridge this gap, this paper provides a comprehensive analysis of the challenges in blockchain PQC migration observed as of December 2025. Unless otherwise stated, this paper focuses on large-scale public, permissionless blockchains with significant economic value and long-lived state, such as Bitcoin and Ethereum. By synthesizing discussions from academia and the blockchain community, this work establishes a system-level taxonomy and an incentive-aware analysis of PQC migration challenges. Rather than proposing new cryptographic primitives or protocol implementations, we provide a roadmap for stakeholders navigating the path toward a quantum-secure future.

The remainder of this paper is organized as follows. Section 2 outlines the fundamental factors obstructing blockchain PQC migration. Achieving crypto agility to enable the seamless replacement of cryptographic algorithms and other primitives is rendered difficult by structural constraints of the decentralized architecture and the inherent nature of blockchains as financial systems. Section 3 enumerates the components requiring migration within the modern blockchains and discuss the technical challenges associated with each. Section 4 analyzes the governance difficulties arising from the intertwined interests of various stakeholders. Finally, Section 5 outlines future research directions that academia should pursue in collaboration with blockchain communities and concludes the paper.

2 Blockchain and Crypto Agility

2.1 Inapplicability of Traditional Cipher Suite Solutions

In general, secure communication protocols such as TLS have an established mechanism wherein the client and server negotiate a cipher suite to dynamically

select and agree upon algorithms and parameters. This modularizes individual algorithms, enabling phased migration and rapid response to novel vulnerabilities. However, in a blockchain, an indefinite number of nodes on a decentralized network must strictly adhere to identical protocol rules to maintain consensus. Nodes that do not comply with pre-defined rules are designed to be rejected and excluded by the network. Consequently, the approach of dynamically negotiating and selecting cryptographic algorithms during node communication and transaction verification as TLS does in cipher suite mechanism is inapplicable.

2.2 Challenges in Software Upgrades in Decentralized Networks

Due to its governance structure without a central authority, software upgrades themselves entail significant difficulty for public blockchains. The fundamental reason lies in the absence of a protocol-level mechanism to identify "honest nodes" within the network and to securely deliver upgrades to them. Indeed, in addressing CVE-2018-17144 [5], a severe inflation vulnerability in Bitcoin, although no attack actually occurred, Bitcoin Core developers were compelled to discreetly contact trusted mining pools and operators individually to request the application of patches before making a public Full Disclosure.

Furthermore, since there is no coercive force exercised by a central administrator, enforcing a simultaneous upgrade across all nodes is structurally difficult. If all nodes do not upgrade simultaneously, there arises a risk of fork due to protocol discrepancies. Moreover, network fragmentation and hash rate fluctuations during the migration process may temporarily degrade the security strength of the entire network, potentially exposing vulnerabilities to threats such as 51% attacks.

2.3 Adverse Incentives and Optimism among Stakeholders

Beyond technical challenges, the economic incentives of the stakeholders pose a critical barrier to proactive migration. Miners and investors tend to act based on short-term economic rationality. Openly discussing the quantum threat casts doubt on blockchain security, risking a decline in asset value. Consequently, stakeholders fearing the depreciation of asset value possess a strong incentive to shun the discussion regarding PQC migration. Furthermore, in formulating the migration process, there is a risk that minimizing the impact on asset value and business profitability will be prioritized over truly secure design, thereby sacrificing security.

Even for end-users, unless attacks by CRQCs become visible, they do not recognize this as an imminent threat and lack a rational incentive to accept migration plans that require high-level technical understanding or additional fee burdens. While optimistic assertions such as "assets are safe as long as verification keys are not exposed on-chain (i.e., strict avoidance of address reuse)" are observed in some quarters, these are rationalizations to postpone fundamental solutions. The short-sighted preference for avoiding risks and costs associated

with migration, coupled with a lack of urgency, constitutes the primary factor stagnating the discussion.

3 Blockchain Complexity and Technical Challenges

Before detailing individual technical challenges, we explicitly distinguish between technical constraints and incentive misalignment. Technical limitations, such as increased signature size or verification cost, do not by themselves prevent migration; rather, their economic and operational consequences reshape stakeholder incentives. In this sense, human factors amplify technical constraints instead of merely coexisting with them.

3.1 Transaction Signatures and Scalability

The most critical technical challenge in PQC migration is the substantial increase in signature and key sizes. This imposes direct and severe implications on blockchain scalability and decentralization.

Key and Signature Size Expansion On the Bitcoin network, a compressed ECDSA verification key occupies 33 bytes, and a signature is approximately 71 bytes; consequently, the total on-chain storage requirement per transaction signature is merely around 104 bytes. In contrast, the PQC signature algorithms currently being standardized by NIST [31] generate verification keys and signatures that are orders of magnitude larger than those of traditional elliptic curve cryptography.

Table 1 presents a comparison of key sizes, signature sizes, and signing / verification costs between various PQC signature algorithms and conventional digital signature algorithms [34]. For instance, CRYSTALS-Dilithium-44 [14], which offers a security level equivalent to AES-128, necessitates a 1312-byte verification key and a 2420-byte signature. Regarding Falcon [30], which is viewed by blockchain communities as a promising alternative algorithm, although it is more compact, it still requires an 897-byte verification key and a 666-byte signature. Unless dramatic compression techniques are discovered in the future, the significant increase in key and signature sizes remains inevitable.

Table 1. Comparison of PQC Signature Algorithms

Signature Algorithm	Verification Key Size	Signature Size	Signing Cost vs. EdDSA	Verification Cost vs. EdDSA
EdDSA	32 bytes	64 bytes	1.0x	1.0x
RSA-2048	272 bytes	256 bytes	∼640x	∼0.35x
CRYSTALS-Dilithium 44	1312 bytes	2420 bytes	∼8x	∼0.9x
FALCON 512	897 bytes	666 bytes	∼24x	∼0.6x
SPHINCS+ 128s	32 bytes	7856 bytes	∼111,000x	∼37x
SPHINCS+ 128f	32 bytes	17088 bytes	∼5,700x	∼99x
SQIsign I	65 bytes	148 bytes	∼2,400x	∼39x

The Trade-off between Throughput and Decentralization The dramatic increase in transaction signature and verification key sizes presents a severe trade-off between network throughput and decentralization. Within the constraints of block capacity, the enlargement of signature sizes reduces the number of transactions stored per block, leading to a significant drop in Transactions Per Second (TPS). While expanding block size offers a countermeasure, it necessitates increased bandwidth and storage capacity for nodes, thereby conflicting with the pursuit of decentralization. For example, Ethereum presupposes the node operation on consumer-grade hardware. The escalating hardware requirements associated with PQC migration act as a barrier to entry for general users, creating a situation where only well-funded organizations can sustain node operation. This inevitably invite network centralization, undermining the blockchain's fundamental value.

Risks of Off-chain Additional Proofs and Single Points of Failure To mitigate the bloat of on-chain data, proposals have been made to store additional data accompanying PQC migration, such as post-quantum zero-knowledge proofs demonstrating ownership of signing keys, in an external storage layer. In the model proposed by Fan et al. [20], only small URIs pointing to proof data stored in an off-chain environment are recorded on the blockchain. While this approach effectively suppresses the increase in transaction data size, it renders the signature verification process heavily dependent on the availability and integrity of external data. This introduces a new single point of failure; should the off-chain storage suffer an attack or failure, there is a risk that transaction verification itself becomes impossible.

An alternative approach is to store additional data on a secondary chain, termed a "support chain" [10, 35]. However, the security of such support chains may be compromised, as they are likely to be maintained by a significantly smaller set of validators compared to the main chain. Furthermore, if participation in maintaining the support chain is limited to nodes possessing sufficient computational power and storage capacity, the increase in operational costs becomes a barrier to entry, meaning the risk of reduced decentralization persists.

3.2 Compromise of Signature Aggregation and Other Components

In major blockchains such as Bitcoin and Ethereum, advanced cryptographic primitives are employed to enhance scalability. However, many of these lack quantum resistance, and their compromise could have a catastrophic impact on system performance.

Signature Aggregation In Ethereum's PoS consensus, BLS signatures facilitate the aggregation of thousands of individual validator signatures into a constant-sized single signature. Furthermore, because verification can be performed at a low cost, they are utilized in the large-scale voting process required for block finalization. However, the security of BLS signatures, much

like ECDSA, relies on the hardness of the discrete logarithm problem on elliptic curves.

As of 2025, PQC signature algorithms and aggregation techniques that offer aggregation efficiency comparable to BLS remain in the research phase [13, 23]. There is a concern that the loss of efficient signature aggregation mechanisms accompanying the PQC migration will become a significant bottleneck in future PoS consensus, leading to lower throughput. It should also be noted that Schnorr signatures, which are adopted for transaction signature aggregation in Bitcoin, similarly lack quantum resistance.

KZG Commitments KZG polynomial commitments are utilized in the Blob commitments within EIP-4844 Proto-Danksharding [9] and in Verkle trees, which are under consideration for introduction in Ethereum's upgrade roadmap [15]. The security of KZG commitments also relies on the discrete logarithm problem on elliptic curves and is not quantum-secure. Therefore, to maintain these functionalities, migration to alternative quantum resistant commitment schemes is imperative. Alternative schemes must be evaluated, balancing trade-offs in proof size and verification time.

3.3 Scaling Solutions

With the advancement of scaling technologies such as Layer 2, PQC migration is required for both off-chain protocols and the interfaces connecting them to the main chain, commonly referred to as "pegs."

ZK-Rollups Many zk-SNARKs adopted by certain projects [36, 41] base their security on the hardness of the discrete logarithm problem, which lacks quantum resistance. While migration to hash-based zk-STARKs is viewed as a promising solution, it necessitates the redesign of proving circuits and verification systems. Furthermore, as zk-STARKs produce larger proof sizes, this transition entails a trade-off involving increased on-chain verification costs.

Optimistic Rollups and Fraud Proofs The security of Optimistic Rollups [1, 32] is maintained via fraud proof mechanisms, wherein claims and counter-claims submitted by participants during the dispute resolution game are authenticated using digital signatures. If these signatures are based on algorithms vulnerable to CRQCs, such as ECDSA, attackers could forge signatures to obstruct legitimate challenges or induce system failure through fabricated challenges.

HTLC (Hashed Timelock Contract) In HTLCs, which serve as the foundation for systems such as Lightning Network [24] and Atomic Swaps [40], if signatures are vulnerable, attackers could derive the signing key following transaction publication and misappropriate funds before the time lock expires. Existing channels and contracts with extended lock times, in particular, are exposed to significant risk due to the difficulty of retroactive modification.

3.4 Smart Contracts and Immutability

The immutability of smart contracts constitutes a core value of the technology, yet simultaneously represents the greatest barrier to PQC migration.

Upgradable Proxy Patterns and Central Administrators Since signature verification logic hardcoded into deployed contracts is immutable, the contract effectively harbors a permanent vulnerability if the underlying algorithm gets compromised. One pragmatic solution involves the adoption of proxy patterns, which separate the proxy holding state variables from the upgradable contract implementing the logic. However, the existence of a central administrator possessing upgrade authority contradicts the "trustlessness" that is the essence of smart contracts. While delegating administrative authority to a DAO is a possible mitigation strategy, the latency of decision-making via decentralized governance models presents a significant challenge. This lack of agility raises concerns regarding the system's ability to respond to highly urgent threats, such as the sudden emergence of CRQCs.

Token Remapping The ownership of tokens, such as NFTs, is typically managed via mappings within smart contracts. In the context of PQC migration, when users transition to new quantum-resistant address systems, the rewriting of records on the contract becomes necessary. In particular, for tokens possessing "non-transferable" functionality, such as Soulbound Tokens, since the transfer of ownership by the user themselves is prohibited, accommodating address changes is impossible without contract-level intervention. Rectifying this necessitates a complex and costly process: taking a snapshot, deploying a new contract, and executing an airdrop to the new quantum-resistant addresses. Furthermore, in the case of projects where developers are no longer active, there exists a risk that appropriate migration measures will not be taken, leaving tokens stranded at the old addresses.

4 Governance Difficulties of Intertwined Interests

4.1 The Challenge of Decentralized Consensus

Blockchain PQC migration extends beyond the mere technical substitution of cryptographic primitives. Since public blockchains lack a central administrator, decision-making and execution of migration are entrusted to the community's autonomous consensus formation. Compared to functional expansions such as the introduction of scaling technologies, reaching consensus on PQC migration is extremely difficult because economic incentives and the motives of stakeholders are intricately intertwined. This section structures this governance challenge by examining currently proposed migration strategies and analyzing the trade-offs between strategies and stakeholder interests.

4.2 Signature Scheme Migration

Baldimtsi et al. [2] has classified the signature migration strategies into four types. We classify them into three categories based on the level of user intervention required: (1) strategies where users keep using only legacy secrets, (2) strategies requiring users to generate new PQC keys in hybrid signature manner without mandating the transfer of funds, and (3) strategies requiring users to execute transactions to transfer assets to a new address system.

Sole Reliance on Legacy Secrets This approach allows users to demonstrate legitimate ownership of an ECDSA signing key even after a compromise. Specifically, it involves generating a post-quantum zero-knowledge proof attesting that the prover possesses the mnemonic seed, such as those defined in BIP-39 [33], which serves as the pre-image for the signing keys. Since the key derivation process relies on cryptographic hash functions, the mnemonic seed itself remains secure against CRQCs. While seemingly a powerful approach that avoids user's key regeneration, it is incompatible with wallet implementations that lack mnemonic seeds, including hardware wallets and non-deterministic legacy wallets. Furthermore, it has been noted in [2] that integration with the hierarchical, scalar-based key derivation mechanisms used in ECDSA is not technically composable.

Hybrid Signatures and Smart Contracts This requires users to generate new PQC keys to sign transactions, often used in combination with existing keys or wrappers.

- *Hybrid Signatures:* Hybrid configurations are currently being considered for PQC implementations in protocols [19]. In particular, a PQ/T (Post-Quantum / Traditional) hybrid configuration enables a phased migration and maintains backward compatibility by attaching both a PQC signature and a conventional ECDSA signature. However, this results in larger transaction signature sizes and increased data storage requirements, as well as higher verification complexity.
- *Protection via Smart Contracts:* Actively discussed within the Ethereum community [17, 18], this method leverages Account Abstraction to introduce arbitrary signature verification logic without L1 protocol changes. This flexibility allows for the adoption of pure PQC signatures or hybrid configurations, for instance, by including a PQC key as one of the authorized signers alongside the legacy key. However, bundlers transmitting transactions may still rely on ECDSA, and the contract upgrade mechanism itself may introduce new security risks.

Migrating Assets to New Address System This is the most radical approach, requiring users to actively create new PQC signing keys and execute on-chain transactions to transfer assets from old addresses to new quantum-resistant addresses(e.g. BIP-360 [4]). As exemplified by the proposal by Lopp et

al. [27], this method progressively invalidates conventional signatures at the protocol level, ultimately forcing the acceptance of only those transactions signed with PQC algorithms. While this is the most definitive method of migration, it necessitates the active participation of the users making a new wallet and transferring the asset within the migration period.

The Analysis of Trade-off and Incentive Confliction The comparative analysis of these strategies reveals a tension between the operational burden placed on end users and the overhead imposed on the blockchain protocol. The "Sole Reliance on Legacy Secrets" approach minimizes the operational costs for users. Since it does not mandate new key generation or on-chain transactions, users can avoid immediate transaction fees and the risks associated with managing new post-quantum keys. However, this simplicity for the user comes at the cost of implementation complexity and computational overhead for the network. Verifying post-quantum zero-knowledge proofs is computationally intensive and exacerbates block size constraints significantly, even when compared to standard PQC signatures[7]. Consequently, while this strategy may be preferred by users, it is likely to face opposition from node operators and developers who must bear the increased validation costs and technical debt.

Conversely, "Migrating to New Addresses" ensures minimal protocol overhead and technical simplicity by utilizing standardized PQC signatures. However, this approach inevitably imposes transaction fees and transfer process management on individual users. These user burdens conflict with the necessity of active user participation, as users are primarily motivated to avoid financial costs and security risks during the migration process. Thus, while favored by developers and node operators for its technical purity, it is difficult to expect end users to voluntarily engage in the migration without any incentive mechanisms.

The "Hybrid Signatures and Smart Contracts" approach represents a middle ground. Hybrid schemes or Account Abstraction wrappers often leverage existing transaction structures. While some initialization such as PQ key generation and key binding may be required, it does not necessarily mandate the movement of funds. Although it incurs certain redundancy and implementation complexity, it enables a phased transition that balances user burden with protocol overhead, potentially mitigating the extreme trade-offs observed in the other two categories.

Conflicting economic incentives between users and miners regarding migration costs are also noteworthy. While general users seek to avoid the transaction fees associated with migration, miners may welcome them as a revenue opportunity. Especially if a strict migration deadline were set and users rushed to move funds within that timeframe, transaction fees would soar. This creates a structure of conflicting interests where miners enjoy enormous profits while users are compelled to bear a cost burden.

[7] ZK-STARK aggregation may mitigate this, though it entails trade-offs such as significant computational overhead for the prover.

4.3 Legacy and Dormant Asset Migration

The Dilemma of Un-migrated Assets Blockchains contain vast quantities of dormant assets that have not been accessed for extended periods due to reasons such as the loss of signing keys or the death of owners [8]. These include assets in legacy P2PK (Pay-to-Public-Key) addresses, typified by Satoshi Nakamoto's coins, where verification keys are exposed on the blockchain, making them vulnerable to immediate CRQC attacks. Regarding the assets that are expected to remain un-migrated after the transition period, including dormant assets, the following approaches are being discussed:

- **Mandatory Migration and Burning:** This strategy involves establishing a transition window to urge users to move assets to quantum-resistant addresses, subsequently invalidating spends from legacy addresses after the deadline to prevent theft by attackers with CRQCs. While this prevents the misappropriation of un-migrated assets, it deprives wallet owners of their property rights if they are unable to perform the migration, thereby violating the fundamental blockchain principles of censorship resistance and self-sovereignty. The Bitcoin migration proposal by Lopp et al. [27] falls into this category [9].
- **Maintaining Backward Compatibility:** Conversely, this strategy keeps un-migrated assets in legacy addresses valid even after the emergence of CRQCs. This approach requires only the implementation of a new PQC signature scheme as an optional alternative. Although this avoids freezing assets that were not migrated during the transition period, it allows attackers with CRQCs to seize those assets and liquidate freely. This is tantamount to a redistribution of wealth from those who are unaware of quantum threats to those who have won the technological race to acquire CRQCs [26].

The Analysis of Trade-off and Impact The dilemma of un-migrated assets presents a profound trade-off between the preservation of fundamental blockchain principles and the mitigation of systemic risks posed by CRQCs. Furthermore, this dilemma necessitates an analysis of how economic impacts are distributed, although predicting these outcomes remains highly uncertain.

The "Mandatory Migration" strategy effectively prevents attacks but compromises core principles such as censorship resistance, and consequently faces significant opposition from the community [6]. This approach imposes the economic burden and risk of loss directly onto dormant asset holders. There is also a risk that asset values drop if the market perceives that the community has compromised the blockchain ethos. Stakeholders who prioritize asset value

[8] For Bitcoin, it is estimated 20% of total liquidity [11].

[9] Although this proposal includes a phase to explore recovery mechanisms for frozen assets such as the use of post-quantum zero-knowledge proofs of seed phrases, it remains incompatible with older wallet standards that lack seed phrases.

preservation, such as investors, may support this plan if they believe it minimizes the overall impact on market price. Conversely, blockchain enthusiasts who strictly adhere to the core ethos are likely to oppose it.

On the other hand, the "Maintaining Backward Compatibility" strategy preserves core principles but leaves the critical vulnerabilities. The continued validity of exposed legacy keys allows adversaries to steal funds, and the potential influx of stolen assets into the market could cause hyperinflation and erode confidence in the crypto assets, destabilizing the asset price. Moreover, the network itself risks losing credibility due to successful attacks, potentially leading to a significant decline in asset value. Unlike the mandatory approach, this strategy does not concentrate the financial burden on specific stakeholders; instead, it socializes the risk, requiring the entire ecosystem to tolerate the potential fallout. It is important to note that the financial impact of either scenario is unpredictable, and this uncertainty further fragments the community discussion.

The Hourglass Model To address this dilemma and balance between the extremes of burning and liquidation, the "Hourglass" proposal [3] has been introduced as an incentive-compatible mitigation strategy. It limits the number of spends from compromised addresses to exactly one per block following the migration period. This scarcity of slots creates a competitive fee market; if multiple actors possess quantum capabilities, they are forced to bid against each other to include transactions from compromised addresses (e.g. P2PK) in the single available slot. This mechanism is claimed to have two effects:

- **Miner Compensation:** The bidding war transfers a significant portion of the recovered value from attackers to miners in the form of high transaction fees, aligning with miner incentives.
- **Early Detection:** It provides a first-mover advantage for attackers to show their abilities early rather than hiding to accumulate more signing keys. This allows the ecosystem to detect the onset of CRQCs based on the spike in transaction fees for the slot.

Conflict with Regulatory Authorities A potential conflict exists between the demands of regulatory authorities and the decisions made by the blockchain community. If the community rejects the protection of vulnerable assets via mandatory migration, it implies a disregard for the threat of large-scale asset theft, which directly contradicts the objectives of Anti-Money Laundering and Countering the Financing of Terrorism measures. In particular, should dormant assets be stolen on a massive scale by state actors or terrorist organizations, there is a concern that this could escalate into a national security risk, unmanageable within existing financial regulatory frameworks.

While there is a counter-argument positing that "attackers will not profit because market prices will crash if the attack is exposed," this assertion remains questionable. Following the realization of CRQCs, if asset movements occur from vulnerable addresses, it is difficult to immediately distinguish whether these are

transfers by legitimate owners or thefts by attackers. Furthermore, attackers may not necessarily steal assets from a large number of addresses simultaneously; there is a possibility they will proceed with attacks gradually and clandestinely to avoid detection. In the latter case, specifically, the theft becomes indistinguishable from signing key leakage due to conventional attacks such as social engineering or phishing. Additionally, methods to obfuscate transaction graphs, such as swaps through DeFi protocols and use of mixing services, remain effective. Thus, there is a certain probability that attackers will successfully launder and cash out assets before the market reacts and collapses.

4.4 Systemic and Platform Migration

The migration methods discussed in the previous sections primarily focused on replacing transaction signature algorithms at the protocol layer. In this section, we examine migration strategies at higher layers and broader execution models.

- **Reactive Emergency Measures:** Within the Ethereum community, several proposals explore reactive strategies that focus on immediate response over proactive upgrades. For instance, smart contracts known as "quantum canaries" have been proposed [16]. In this model, a cryptographic puzzle (the canary) is deployed with a massive bounty; once solved, it signals a CRQC breakthrough and triggers an automatic migration process. Similarly, Buterin [17] suggests a hard fork strategy to "revert all blocks after the first block where it is clear that large-scale theft is happening" as an initial step. While these approaches minimize upfront costs, they may require emergency implementation, potentially leading to significant confusion within the community.
- **Proxy Solutions:** This strategy involves using an intermediate layer to protect assets without altering the base protocol immediately. A prime example is the "Solana Winternitz Vault" implemented by the Solana community [25], which utilizes quantum-resistant Winternitz One-Time Signatures (WOTS). In this model, users deposit assets into a "Vault" smart contract that can only be unlocked via a WOTS signature. This solution is offered as an opt-in feature for risk-conscious users seeking long-term protection and can be deployed immediately. However, there is a risk that only a subset of users will utilize it, while the majority of general users may continue to postpone action, leaving a significant portion of assets vulnerable.
- **Platform Migration:** This strategy involves abandoning legacy chains in favor of blockchains designed from the ground up to be quantum-safe, such as Cellframe [12] and Quantum Resistant Ledger (QRL) [39]. For example, QRL recently introduced the "Zond" update [38], making it EVM-compatible, and positions itself as a safe haven in an uncertain world of quantum computing. If the upgrade of existing blockchain infrastructure proves unsuccessful and a shift to these alternative platforms happens, it would necessitate the forfeiture of the ecosystem's vast intellectual memory and collective engineering wisdom.

4.5 The Analysis of Migration Trilemma

In PQC migration strategies, we observe a trade-off among "Decentralization," "Security," and "Overhead." For instance, approaches such as reactive rollbacks or delegating migration to trusted proxies allow for rapid and relatively simple implementation. However, because they involve top-down decision-making or reliance on a specific entity or codebase, they compromise decentralization by introducing a Single Point of Failure. Conversely, naive strategies that depend on the autonomous decision-making of individual stakeholders face significant challenges in achieving consensus, as previously discussed. Consequently, this invites the risk of chain splits via hard forks and the fragmentation of hash rate, potentially degrading the security of the entire network. Lastly, approaches designed to minimize user burden during migration, in an effort to reconcile decentralization and security, inevitably incur overhead. This includes larger data sizes and verification costs stemming from the incorporation of additional cryptographic mechanisms, as discussed in 4.2.

5 The Future of Blockchain PQC Migration

5.1 Defining PQC Migration Requirements

Ideal Cryptographic Requirements Meng et al. [28] define the requirements for blockchain cryptographic migration schemes as follows:

- **Correctness:** If blocks and signed transactions are generated following the scheme, they must always pass hash and signature verification, thereby maintaining blockchain validity.
- **Long-term Integrity:** Regardless of the lifetime of the underlying cryptographic algorithm, no polynomial-time adversary can compromise the scheme's integrity. Specifically, no entity can claim the existence of non-existent data or tamper with data within any block.
- **Long-term Unforgeability:** Regardless of the lifetime of the underlying cryptographic algorithm, no polynomial-time adversary can forge signatures. No entity should be able to forge a signature based solely on knowledge of the public key.

Practical Requirements and Necessary Relaxation To achieve a migration that is acceptable to the community and sustainable for the ecosystem, it is essential to consider practical requirements that account for operational realities and human factors. While these requirements are founded upon the ideal requirements above, practical implementation may necessitate their relaxation. Specifically, the following requirements are conceivable, though they are not exhaustive and should be scrutinized through future discourse.

1. **Security During the Transition Phase:** Even if computational costs or data size overheads are incurred during the migration process, the blockchain's

security level (e.g., hash rate or validator count) must not be significantly compromised. Furthermore, if an unexpected attack occur during the transition period, the system must maintain a state where remedial measures or mitigation strategies can be applied immediately, without waiting for the migration process to complete.

2. **Remedial Mechanisms for Dormant Assets:** To maintain the blockchain's function as a cash system for economic activity, some form of remedial measure is required for dormant assets. For instance, even if protocol-level migration is difficult, it is desirable to provide a means to prove legitimate ownership off-chain and recover assets. Note that providing such remedial mechanisms implies that strict "Long-term Unforgeability" may not be upheld for legacy addresses if users fail to act.
3. **Comprehensive Migration Beyond Transaction Signatures:** As a prerequisite, it is crucial to define "what constitutes a quantum-secure blockchain" by considering migration requirements specific to each chain in addition to general requirements. Discussions regarding migration tend to focus on the replacement of transaction signature algorithms; however, simply migrating signature algorithms does not render the entire system secure if vulnerabilities remain in other components. PQC compliance is imperative across the full blockchain protocol stack, including consensus mechanisms and smart contract deployment and execution environments.

5.2 Designing Balanced and Incentive-Aligned Migration Strategies

Apart from technical challenges, the game-theoretic governance problem of motivating the majority of stakeholders to actively participate in the migration demands careful consideration. Addressing this requires the rigorous scrutiny of stakeholder interests, the consideration of balanced solutions, and the design of incentives for early PQC migration. For example, introducing mechanisms such as fee discounts or priority processing in the mempool, as proposed in [7], warrants serious consideration. Such measures aim to create a situation where it is "economically more rational for users and miners to complete migration early rather than delay it."

5.3 Practicing Crypto-Agility

The greatest lesson learned from today's PQC migration discussions is the fact that the current blockchain ecosystem has not sufficiently anticipated the compromise of cryptographic primitives. As a result, it significantly lacks crypto-agility. Recognizing that all cryptographic primitives within blockchains possess a finite security lifespan, crypto-agility must be prioritized as a fundamental design principle. The PQC migration should not be viewed merely as a reactive, one-time transition, but rather as an architectural improvement that ensures long-term system resilience. By practicing crypto-agility, we should pursue the construction of a sustainable blockchain infrastructure capable of adapting to future unknown threats without destructive changes or ecosystem fragmentation.

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